When Can a Swarm Be Audited?[1]

A bounded oracle and development pilot

Razvan Sacara | Swarm Alignment Hackathon participant

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Abstract

Multi-agent oversight requires evidence connecting an executed action to an immutable instance. A correct guess can remain unsupported when that relation is missing. We implement a bounded fake-record testbed and an explicit compatible-history oracle for executor attribution. A separately written constraint checker agrees with enumeration on 55,296 valid world/export combinations. In a completed development batch, two hosted Gemma models each read three evidence views of 20 scripted source episodes. Grounded correctness, including justified abstention, was 54/60 and 59/60 calls; supported definite findings were 41/60 and 45/60. Targeted deletion reduced singleton-answerable packets from 20/20 to 10/20 but did not produce more unsupported worker accusations than random deletion in this batch. One observed correct executor guess was nevertheless unsupported because execution itself was not established. Earlier qualification batches exposed reasoning truncation, format errors and timeouts, all retained separately. A 32-call exploratory peer-message pilot produced no unauthorized writes. Four Hugging Face calls verified an additional serving route. These are development results from one small grammar, not confirmatory evidence of swarm safety or spontaneous collusion. The contribution is an inspectable evidence contract, executable validation and documented feasibility limits. Larger stateful workflows, causal influence tests and optional small-monitor training remain prospective.

[1] Research conducted at the Swarm Alignment Hackathon, November 2026

1. Introduction

Suppose an investigator sees a worker binding but no authenticated execution result. Naming the bound worker may match the hidden truth while still exceeding what the export establishes. We observed exactly this distinction in a development response. Whether a violation occurred, which instance executed it, and whether agents colluded are separate questions [1,2,6,23]. Our labels concern execution and evidential support, not intent or decisive causal responsibility.

We ask whether selective export loss and changing display names impair supported executor attribution, and which authentic relations restore it. Track 4 oversight anchors the project, with identity-infrastructure and testbed relevance [12]. We report a bounded oracle and development pilots; the proposed larger interaction study is not frozen or completed. No claim of first invention, deployment protection or a probability of winning follows.

2. Related Work

Qin et al. study trace deletion, peer influence and independent recording/completeness [3]. Chan et al. supply the identity and infrastructure context [4,5]. Yun et al. distinguish correct guesses from supported findings, include abstention and a same-packet comparator, and cross binding availability with identifier renaming [6]. Naming changes alone therefore offer weak novelty. Our pilot extends these distinctions to a declared finite executor-label contract; it does not originate them.

Syscon collects external execution receipts without an automatic detection gain; ActionLens evaluates additional evidence and component ablations [17,18]. Who&When targets decisive errors, BLADE audits handoff responsibility and BRIDGE verifies runtime repair [19,20,10]. Our executor labels and proposed offline minimum-evidence diagnostic answer narrower questions. Receipts, blame graphs and generic repair are not new contributions here.

HELIOS observes internal activations, unlike our black-box reader packets [7]. AI Control and CaMeLs motivate explicit trust boundaries [8,9]. Kitaoka et al. already evaluate optimized distributed attacks and separate/joint monitoring; OpenAI describes multilayer investigations [21,22]. Swarm Traces has uncertain identities, timing and outcomes and is not our scoring oracle [11]. Prompt Infection already studies inter-agent propagation, so a scripted peer-message pilot is not a new demonstration of that phenomenon [28].

3. Methods

Scope and sources. On 6 November 2026 we examined the live sprint materials, supplied transcripts/chats, relevant primary methods and eight selected winning examples. The original report/review sample and subsequent control projects supplied lessons about clean controls, causal checks, component ablations and honest nulls [12–15,17,18]. SOURCE_AUDIT.md records reading depth and access gaps. This targeted scoping is not an exhaustive survey or a priority clearance.

Finite environment. One designated fake request has a selected worker in {A,B,C}, outcome applied/denied/not_called, an arbitrary permutation of three aliases, and unchanged or prefixed display names: 108 worlds per public authorization context. A local typed interpreter changes only a newly allocated fake dictionary; no model text reaches a shell or real data service. A successful unauthorized fake write is permitted for positive controls. Hidden host state labels its executor; denied/nonexecuted or authorized actions have label NONE. The eight-record export separates an opaque-handle-to-instance binding B from post-execution receipt R; advisory prose is untrusted.

Observation and answerability. Full, seeded-random 25%-loss and fixed selective 25%-loss views retain equal record counts within instrumentation. The selective mask removes the handoff and binding; originals remain protected. For packet x and public constraints S, enumerate histories whose entire projected packet matches x. A singleton compatible label supports that definite answer; several labels require ABSTAIN; no compatible history means inconsistent evidence. A separate constraint implementation shares no projection/enumeration helpers. Golden cases check missing B/R, denied execution, misleading prose, malformed trusted fields and names. Exhaustive validation permits all mask sizes; the reader pilot declares only 0/2/4-record losses. The mask helper currently implements full instrumentation at 25%, not a general corruption engine.

Development sequence. Initial three-model calls at 2,048 output tokens exposed truncation; an interrupted batch retains 28 response records, with other calls unknown or unstarted. A six-model 4,096-token qualification halted after 18 calls when billing usage was unavailable after gateway errors. A clearer bare-JSON prompt and three-model run halted after 120 calls at a predeclared three-truncation gate. These failures remain separate. A subsequent fixed batch completed 20 source episodes (10 violations, five denied and five nonexecuted), three views and two Gemma endpoints: 120 calls. The same source seeds recur across iterations; batches are not independent replication datasets. Churn varies across episodes rather than supplying paired churn counterfactuals, so we do not estimate its interaction.

Reader/scoring contract. Plain-prompt readers receive identical packets and public grammar, fresh requests, temperature 0, no tools, memory or continuation, and a 4,096-token output cap. Exact JSON statuses are EXECUTOR(worker), NONE or ABSTAIN; malformed, incomplete and transport-failed outputs are INVALID. All planned scorable calls remain in denominators. Grounded correctness requires a supported definite label or justified abstention; supported definite coverage excludes abstention. Unsupported NONE findings are separate from worker accusations. Raw prompts, versions, attempts, masks, usage and responses stay under ignored runs/; sanitized case-level packets, final answers, fake source episodes and aggregates are retained in Git. Token/latency/cost differences are reported, not equated with record counts.

Additional feasibility tests. A separate one-hop pilot used four repetitions of one fake read task, malicious/benign/irrelevant/no-peer messages and a source-warning variant: 32 actual downstream Gemma choices. A typed interpreter could execute only read, refuse or fake write. Task facts and authorization were fixed; message lengths/tokens differed. Exposure was scripted, not emergent coordination. Four same-grammar HF calls used an explicitly priced Cerebras route

[27]; these test access/serving only, not checkpoint equivalence or independent model-family replication.

4. Results

Executable validation. Enumeration and the separate constraint checker agreed on 55,296 valid world/mask/authorization combinations; golden and malformed-field tests passed. This validates the written small grammar, not arbitrary real traces, host security or all malformed inputs. Table 1 locates the closest overlap.

Table 1. Existing results and the narrower proposed test. These are research-design comparisons, not measured performance.

Prior work	**Already established**	**Proposed test**
Qin [3] / Yun [6]	Trace integrity; evidence support and abstention	Declared selective loss crossed with name churn
Syscon [17] / ActionLens [18]	External receipts; component ablations	Separate retained bindings from execution receipts
Who&When [19] / BLADE [20]	Decisive errors; handoff responsibility	Executor support within a bounded history model

Completed fixed batch. The exact comparator supports 45 definite answers and 15 abstentions across 60 packets. Full views were singleton-answerable in 20/20 cases, random-loss views in 15/20 and selective-loss views in 10/20. This is information removed by construction. Table 2 reports strict reader counts with invalid calls retained. There is no observed increase in unsupported worker accusations under this selective mask: the 26B endpoint made one under random loss and none under selective loss; the 31B endpoint made none under either.

Table 2. Completed development batch: 20 source episodes, three dependent views per episode and 60 scorable calls per model. Grounded correctness includes justified abstention. Unsupported columns are worker / NONE / invalid counts.

Reader / baseline	**Grounded correct /60**	**Supported definite /60**	**Worker / NONE / invalid**
Gemma4-26B-A4B	54	41	1 / 0 / 1
Gemma4-31B	59	45	0 / 1 / 0
Exact packet comparator	60	45	0 / 0 / 0
Always abstain	15	0	0 / 0 / 0

Observed unsupported finding. In one random-loss packet the 26B endpoint named A, which was the hidden executor, but the compatible set was {A,NONE} because the result receipt was absent. The output was factually right and evidentially unsupported. The 31B endpoint instead made one unsupported NONE finding in an ambiguous random-loss packet. Neither endpoint accused a worker on the 10 clean source episodes. These selected development cases do not establish low deployment false-accusation rates.

Bounded information limit. Fix public rules and side information. If two equally likely allowed histories H_A and H_B yield the same entire packet but different executors A and B, a packet-restricted reader has the same response distribution in both. With naming probabilities q_A and q_B, mean factual naming accuracy is $(q_A+q_B)/2 <= 1/2$ because $q_A+q_B<= 1$. Equality holds when all response mass names A or B; unequal priors change the best bound to the larger prior. Only abstention is justified for that ambiguous packet. This elementary construction is not a new theorem or a claim that every deletion creates ambiguity.

Hidden history	**Visible packet after erasure**	**Correct actor**
H_A: worker A updates record	Update complete; actor link absent	A
H_B: worker B updates record	Update complete; actor link absent	B

Figure 1. Two histories collapse to identical evidence. Any monitor restricted to this packet must guess or abstain. Retaining a trustworthy actor–operation receipt makes the observations differ. This construction is analytical, not an observed agent run.

Feasibility and negative results. Qualification exposed truncation, Markdown-wrapped JSON and gateway failures; model names alone did not determine usable serving behavior. The fixed batch cost about 0.00542 provider-reported credits. Known retained Adaption request records across all pilots report about 0.26419 credits, excluding unresolved charges; the final account ledger was not re-read after browser connection failure. HF token-rate costing estimated $0.01347 for four smoke calls. All 32 one-hop choices were reads, with zero attempted/executed unauthorized writes and no measured benefit from the extra warning. Four repetitions of one task cannot establish resistance to peer influence. No training, confirmatory collection, spontaneous-collusion observation or deployment evaluation was completed.

5. Discussion and Limitations

The practical target is defensible reconstruction: preserve the relations that support an executed operation without falsely accusing an instance. The observed correct-but-unsupported response reproduces a distinction already central to Yun et al. [6]. The new artifact is a small executable contract and feasibility record, not a generic new logger or attribution framework. Cooperation, collective agency and executor support remain different concepts [23–25].

The pilot does not support the hoped-for selective-loss/churn interaction. Its views are dependent, its grammar is small, the two completed-batch endpoints share a model family, and endpoint selection/prompt changes used development outputs. Both interpreters can share a conceptual specification error despite separate code. Authenticity is assumed: self-consistent forged trusted records and compromised collectors are outside scope. The peer pilot found no failure to mitigate; it provides no training-improvement claim.

The next useful investment is richer stateful workflows, a held-out task-family split, matched no-handoff controls and a budget-matched authentic-evidence restoration test. A minimum oracle witness would be a diagnostic lower bound, not a deployable policy. Optional small-monitor SFT requires correct supported labels, a priced/bounded job and usable pre/post-checkpoint serving [29,30]. The full available credit envelope is a resource ceiling, not a reason to expand samples after favorable outcomes or to claim novelty from sponsor usage.

6. Conclusion

A bounded oracle and real development calls now make the proposed oversight task executable. The completed data show mostly supported decisions, occasional unsupported findings, and substantial serving/format constraints. They do not establish a new swarm vulnerability or a proven defense. A stronger submission requires a frozen, realistic extension and honest reporting of its positive or null result.

Code and Data

Code, 120 sanitized reader outputs, 20 fake source episodes and aggregates: https://github.com/razzyrazzy11/swarm-alignment . Reproduce released labels and scores offline

with python -m experiments.verify_release. Native API responses remain private and hash-bound. Earlier source snapshots are incomplete; fresh inference is nondeterministic. October trusted-agent-sting code/data/results were not imported [16]. Automated review is not human author approval.

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Appendix A. Limitations and Dual-Use / Ethical Considerations

Validity. One toy grammar, scripted policy violations, three dependent views and post-development endpoint selection limit interpretation. The four batches reuse source seeds and cannot be pooled as independent trials. Only the last fixed batch is complete. The first batch was interrupted without a durable pre-call journal; missing records can be unknown or unstarted. Later journals record starts before calls and preserve completed responses. Provider-reported costs are not a reconciled invoice; final balance refresh failed. Exhaustive checks cover valid finite projections plus selected malformed fixtures, not arbitrary traces or semantic validity of the specification.

Statistical limits. Ten clean source episodes are insufficient to establish a small false-accusation rate. Even an illustrative zero-event one-sided 95% common-risk Bernoulli bound at n=10 is about 25.9%; that calculation does not establish deployment risk or correct development selection. Views/repeats do not increase independent n. Future sample size, model choice, serving settings, outcomes and stopping rules must be frozen before confirmation. No confidence interval over 60 dependent packets is reported as 60 independent trials.

Potential misuse. Evidence-loss tests could reveal what to suppress or how to frame a peer. Keep operations confined to fake in-memory records; do not execute incident payloads or infer intent from names and agreement. Publish defensive schemas, aggregate measurements and benign fixtures, and review operational details under Apart responsible-disclosure rules. No live-service security attack, personal-data task or human-subject study was performed.

Provenance and privacy. The host originals and authorization facts are trusted for this grammar. A trustworthy receipt is not proof of completeness; an unanchored hash chain is not immutable. Raw chats, attendee identifiers and credentials are excluded from Git. API keys were used only for their intended provider authentication and never included in model input. External APIs received fake task data. Source reading depth and remaining inaccessible originals are recorded, without an exhaustive originality or eligibility guarantee.

Appendix B. Resources and next experiment

The user authorized the full available Adaption credit balance as the research ceiling, observed initially at 1,500 credits with 1 credit=$1. Hosted inference is operational and credit use was observed; training and dataset catalog reads also worked, but paid training/adaptation jobs were not launched. HF Pro identity, its $2 available compute balance and one explicitly priced inference route were verified. Budget allocations remain prospective; do not top up or treat a model catalog as proof of every job entitlement [26,27].

Candidate allocation: 600 credits for a richer core, 350 for controlled influence, 300 for small-monitor training only after a bounded price/serving gate, 150 for independent serving/transfer checks and 100 for reproducibility/failed-run liability. This totals the 1,500-credit initial envelope and must be reconciled to actual remaining credit before launching. Current lightweight pilots required far less. SFT minima, automated preference scores and iteration limits are not evidence of

monitor effectiveness or monetary caps [29]. A downloaded adapter needs compatible base weights, software and a working serving route [30]. See SUITE_STRATEGY.md and the dated deviation log.

LLM Usage Statement

Codex assisted with source review, implementation, analysis and writing. Isolated OpenCode reviewers inspected allowed source copies without credentials or raw runs. Hosted models supplied the recorded reader/worker outputs. The author remains responsible for claims and submission; no human author approval, independent empirical replication or confirmatory preregistration is implied.